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Project Title: Predicting Compressive Strength of Concrete Through Non-linear Regression Methods

Problem Statement

Concrete is one of the most frequently used materials in building the world around us and is extensively studied regarding the various applications that it can be used in. Concrete is composed primarily of cement, water, coarse aggregate, and fine aggregate. Coarse aggregate is typically a form of crushed stone and fine aggregate is typically sand. [1] These materials are combined in various amounts as a mixture and can modulate the different physical properties of the final material. Given this information, it would be important for concrete manufacturers to be able to predict the physical properties of the final product from the input variables i.e. the composition and age of the material after it cures into hardened concrete. One of the most important physical properties of concrete is the compressive strength of the material as it needs to be able to withstand a significant load without failing and if manufacturers have a suitable model for predicting the compressive strength of the material, then they can work to tailor the composition to the application. [1] In some cases, this predictive model would be able to not only save money by making concrete more effectively with less waste through off-quality product, but it may also help to save lives when the application has that level of severity and with age of the concrete as another predictor in the model it will also be possible to predict when concrete may need to be replaced after loss of required level of compressive strength.

Data Source

The data used in this research project comes from the UC Irvine Machine Learning Repository. The data set contains 1030 instances with 9 columns containing 8 quantitative feature variables and 1 quantitative response variable [2]. The 7 feature variables regarding concrete composition are the cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate all in units of kg/m^3 meaning the kilograms of each specific material contained inside a cubic meter of concrete. The other feature variable is age of the concrete, and it is measured in days regarding the life of the concrete. The response variable is the concrete compressive strength, and it is measured in MPa or megapascals, which is essentially a unit of pressure that is equivalent to force per unit of area. From the data source Concrete_Readme.txt, "The actual concrete compressive strength (MPa) for a given mixture under a specific age (days) was determined from laboratory. Data is in raw form (not scaled)." [2] It can also be noted that the dataset contains no missing values after running a simple "df.isnull().values.any()" check on the data frame. Below in table 1 is the summary statistics for the dataset. The summary statistics of the values are sensible and do not seem to contain any outliers from initial observations, but more work will be done in the exploratory data analysis section of the report.

Table 1 – Summary Statistics for the Concrete Compressive Strength Dataset

Value	Cement (kg/m^3)	Blast Furnace Slag (kg/m^3)	Fly Ash (kg/m^3)	Water (kg/m^3)	Super-Plasticizer (kg/m^3)	Coarse Aggregate (kg/m^3)	Fine Aggregate (kg/m^3)	Age (days)	Concrete Compressive Strength (MPa)
Count	1030	1030	1030	1030	1030	1030	1030	1030	1030
Mean	281.2	73.9	54.2	181.6	6.2	972.9	773.6	45.7	35.8
Std.	104.5	86.3	63.9	21.4	5.9	77.8	80.2	63.2	16.7
Min.	102.0	0	0	121.8	0	801.0	594.0	1.0	2.3
25%	192.4	0	0	164.9	0	932.0	730.9	7.0	23.7
50%	272.9	22.0	0	185.0	6.4	968.0	779.5	28.0	34.4
75%	350.0	142.95	118.3	192.0	10.2	1029.4	824.0	56.0	46.1
Max.	540.0	359.4	200.1	247.0	32.2	1145.0	992.9	365.0	82.6

Methodology

In this report, initial work will be done to understand the data via exploratory data analysis, where correlation plots will be analyzed to understand how variables interact and whether there are positive or negative relationships with the dependent variables, but also the strength of the correlation between them. The data will then be scaled so that the models have uniform data being fed into the analysis. A new column called 'Water to Cement Ratio' will be created to show how the relationship between amount of water added vs. amount of cement can impact the overall compressive strength of concrete. This addition was inspired from research into the technical details surrounding this topic, where we found that too much or too little water can affect the final properties of the material. [3]

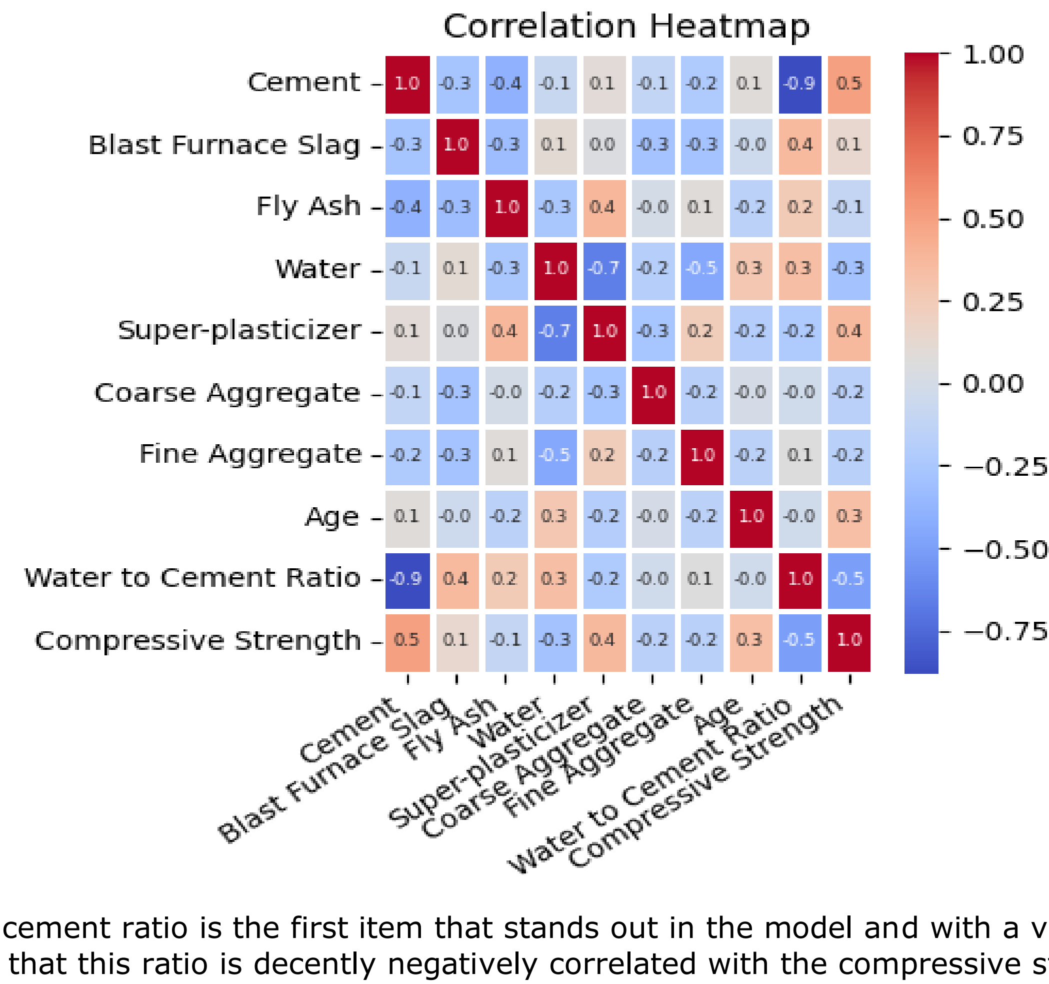
Secondly, linear regression models will be trained on the data so that a baseline for performance can be developed. If any of our non-linear models perform worse than the linear regression model, then it will be obvious that those models should be removed from consideration moving forward. Cross validation will be used to effectively tune the hyperparameters of the models and ensure that the best performance on average can be retrieved from the model.

Finally, non-linear regression models will be trained on the data to be able to predict the response variable. This analysis will look at several different types of models and algorithms including polynomial regression, where polynomial features with a degree of 2 will be evaluated on a linear model. Regularization with lasso regression where we perform 5-fold cross validation and regularization with elastic net regression where we also perform 5-fold cross validation will be performed. These two models will help us understand how terms can be minimized according to the L1 norm penalty, L2 norm penalty, or combination of both applied to the model depending on the type of model. These techniques can be used as variable or feature selection tools since methods like lasso will take certain terms to zero if the model sets it at that. Support vector regression that uses a kernel approximation will be evaluated using a radial basis function as the kernel. It works similarly to support vector machines where a hyperplane can be fit between data points to minimize error. Random forest regression using an ensemble of decision tree regressors on various permutations of the samples and averages performance across the samples. Cross validation will be used to effectively tune the hyperparameters of the models and ensure that the best performance on average can be retrieved from the model. The reason for using non-linear regression models is it is believed that the relationship between the feature variables and the response variable is highly non-linear. It could be reasonably assumed that as the age of the concrete increases, we may see a decrease in the compressive strength of the concrete and this relationship has the potential for non-linearity that may present itself a polynomial curve with degree 2.

Evaluation and Final Results

The information from the exploratory data analysis is the first place to begin the evaluation of the results, since understanding the interactions between this data will guide the understanding and help to further conclusions later.

Figure 1 – Correlation Heatmap of Independent and Dependent Variables



The water to cement ratio is the first item that stands out in the model and with a value of -0.5 this indicates that this ratio is decently negatively correlated with the compressive strength of concrete. This means that as the amount of water in the water to cement mixture increases the compressive strength of the concrete decreases, which is intuitive, but is important to capture in the model. The other variable that has a decent correlation with the compressive strength of concrete is the cement. This indicates that as the amount of cement in the mixture increases the compressive strength of the concrete will increase, as well. The only other variable's worth mentioning in the heatmap are the superplasticizer indicating that as the superplasticizer increases there is a positive relationship with the compressive strength of the concrete and the age of the concrete where it shows that there is a mild positive correlation between age of the concrete and the compressive strength.

The results of the models will be compared below and the metrics of the models that will be analyzed for performance characterization include R-squared value, mean square error (MSE), and the root mean squared error (RMSE). The analysis will look for higher R-squared values for indicating the percentage of variance in the response variable that the model explains through the feature variables and tells us how well our regression curve fits the given data. A lower RMSE is the goal for a measure of the error in the model's predictive power and is essentially a measure of the average distance between predicted results and the data that the model was built upon. The RMSE helps us to understand the magnitude of the error in the same units as the original data. For the cross-validated models (CV), only the mean of the 5-fold RMSE will be reported to show that either the original model is performing well and returning a similar RMSE or that the original model is far from the cross validated model given that the two RMSE's would be far apart.

Table 2 – Table Showing the Results of the Models Evaluated

Model	MSE	RMSE	R^2
Linear Regression	105.5	10.3	0.613
CV Linear Regression	-	11.1	-
Lasso Regression	106.1	10.3	0.611
Elastic Net Regression	106.1	10.3	0.611
Polynomial Regression	55.1	7.4	0.798
CV Polynomial Regression	-	7.9	-
Support Vector Regression	33.6	5.8	0.877
CV Support Vector Regression	-	6.5	-
Random Forest Regressor	24.8	4.9	0.909
CV Random Forest Regressor	-	5.1	-

Firstly, the linear model is built with all coefficients and the assumption that the relationship between all the independent variables and the dependent variable is linear. The linear model tells us that the RMSE between the model's predictions and the actual data is 10.3 and the R-squared of the data is 0.613 out of 1. This indicates that there is a decent amount of error between the data and the linear model's predictions, while the model only captures approximately 61.3% of the variance contained in the data. This is surprisingly decent performance from a linear model considering that the underlying data is known to be non-linear. This is a good baseline for understanding how the future model's performance improves. Two other linear models were also evaluated to understand variable selection and how that impacts model performance. The first alternative method is lasso regression, and this model includes a penalty term that will shrink some coefficients to zero if they are larger than the term allows. This model returned the following coefficients:

[9.1269551, 7.58562844, 4.19330994, -3.95599053, 1.95436914, -0.06400529, 0.0, 7.08234129, -1.54986345]

This shows that the 7th term in the model was shrunk to zero since it was determined to be a coefficient of low importance in the model. This 7th term is the fine aggregate component of the concrete mixture, and this can be interpreted to mean that fine aggregate has no significant impact on predicting the compressive strength of the concrete. The 6th term is the coarse aggregate component of the concrete mixture, and it is a near zero term so this can be assumed to be the next least impactful component of a concrete mixture in terms of predicting the compressive strength. A linear regression model was then built with these model coefficients, and the metrics were essentially the same as the original linear regression model with an RMSE of 10.3 and an R-squared value of 0.613 out of 1. This indicates that even with the unimportant coefficients removed from the model, the predictive capability of the model is not increased over the original linear regression model.

The elastic net regression model was the other alternative method for linear predictive models and incorporates an L2 norm on top of the L1 norm for penalizing coefficients. It is useful since it incorporates both penalty terms to handle multicollinearity and remove coefficients that are unimportant to the model's performance. The model returned the following coefficients:

[8.82200041, 7.59457579, 4.18737205, -3.90107966, 1.95768154, -0.14593127, 0.02115539, 7.08450936, -1.8828881]

This shows similar behavior to the lasso model where the 6th and 7th terms of the model were near or set to zero. The RMSE for the model is 10.3 and the R-squared value is 0.613 out of 1. This indicates that for this application, there is no significant difference in performance between the lasso regression model and the elastic net regression model, giving further evidence that even when imploring models that penalize for terms that are unimportant, reduce multicollinearity, and prevent overfitting, we still get the same performance as the original linear model.

Secondly, the non-linear regression model's performance will be evaluated. The polynomial regression model with a second-degree polynomial was applied to the data set and the results were improved over the linear model, as expected. The RMSE of the model is 7.4 and the R-squared value is 0.798 out of 1. This shows that the polynomial regression model is an improved model over the linear model since the fit was improved and the explained variance given the second order terms was improved. The cross-validation results of the model show that the results from the original model indicate well-tuned hyperparameters. The support vector regression model was an even further improvement over the polynomial regression model. This is sensible due to the leveraging of the radial basis function kernel, where a non-linear decision boundary is developed to perform regression on data that is not linearly separable. The RMSE of the model is 5.8 and the R-squared value is 0.877 out of 1. This indicates that the relationship between the independent variables and the dependent variables is more complex than just a simple second-order polynomial relationship. The cross-validation results of the model indicate that the original model has decently tuned hyperparameters, but the most significant difference between the original model and the cross-validated model suggest that there is more work to be done in other studies to improve the hyperparameters of the original model.

Finally, the non-linear regression model with the best performance was the random forest regression model. This type of data is good for building random forest regression models since it is not sparse and has minimal categorical variables. The RMSE of the model is 4.9 and the R-squared value is 0.909 out of 1, which indicates that this model has the lowest RMSE value over all the other models evaluated and explains the most variance of all the models. The ensemble methods are effective since they essentially evaluate all the permutations of the data and average the performance to achieve the best fit. The cross validation of the random forest model indicates that the original model has well-tuned hyperparameters.

In conclusion, the data set provided is non-linear and linear methods are not effective to determine the compressive strength of concrete given data for the independent variables. Polynomial regression is a decent model to predict the compressive strength of concrete and is a highly interpretable model that can be easily understood by most. Often in materials science, it is important to understand why variables are affecting the final physical results of a product and the most explainable model is favorable. However, with only 79.8 of the variance explained in the model, one must be wary of relying solely on that. The support vector regression model and the random forest regression model indicate that there are complex relationships going on in predicting the compressive strength of concrete and when accuracy is desired for being able to predict the outcome, these models are more favorable.

Works Cited

[1] “What Is Concrete? Types, Composition & Properties.” *UltraTech Cement*, UltraTech Cement Ltd., 2021, www.ultratechcement.com/for-homebuilders/home-building-explained-single/descriptive-articles/what-is-concrete-types-composition-properties-and-uses. Accessed 12 Oct. 2025.

[2] Yeh, I-Cheng. "Concrete Compressive Strength." UCI Machine Learning Repository, 1998, https://doi.org/10.24432/C5PK67.

[3] “What is the Right Water Cement Ratio”, UltraTech Cement, UltraTech Cement Ltd., 2021,https://www.ultratechcement.com/for-homebuilders/home-building-explained-single/descriptive-articles/how-to-calculate-the-water-cement-ratio Accessed 26 Nov. 2025